

DS HOMEWORK

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1: Sum of Two Matrices

```
#include <stdio.h>

int main() {

    int r, c, i, j;

    printf("Enter rows and columns: ");

    scanf("%d%d", &r, &c);

    int a[r][c], b[r][c], sum[r][c];

    printf("Enter first matrix:\n");

    for( i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");

    for( i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);

    printf("Sum Matrix:\n");

    for( i=0;i<r;i++) {
        for( j=0;j<c;j++) {
            sum[i][j]=a[i][j]+b[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }
```

```
}  
return 0;  
}
```

Output:

Enter rows and columns: 3 3

Enter first matrix:

5 10 15

20 25 30

35 40 45

Enter second matrix:

2 4 6

8 10 12

14 16 18

Sum Matrix:

7 14 21

28 35 42

49 56 63

2: Reverse an Array

```
#include <stdio.h>
```

```
int main() {
```

```
int n,i;
```

```
printf("Enter size: ");
```

```
scanf("%d",&n);
```

```
int a[n]
```

```
printf("Enter elements:\n");
```

```
for( i=0;i<n;i++)
scanf("%d",&a[i]);
}
printf("Reverse Order:\n");
for (i=n-1;i>=0;i--)
printf("%d ",a[i]);
return 0;
```

Output:

Enter size: 6

Enter elements:

12 45 67 89 34 10

Reverse Order:

10 34 89 67 45 12

3: Count Duplicate Elements

```
#include <stdio.h>

int main() {
int n,i,j,count =0;

printf("\n Enter the N value:");

scanf("%d",&n);

int a[n],check[n];

printf("Enter the array Elements :|\n");

for( i=0;i<n;i++) {
}

scanf("%d",&a[i]);

check[i] =0;
```

```

for(i=0;i<n;i++) {
}
if(check[i]) continue;
or(j=i+1;j<n;j++) {
if(a[i]==a[j]) {
check[j] =1;
count++;
}
}
}

printf("Duplicate Elements = %d",count);

return 0;

```

Output:

Enter the N value: 7

Enter the array Elements :

10 20 10 30 20 10 40

Duplicate Elements = 3

4: Print Unique Elements

```

#include <stdio.h>

int main() {
int n,i,j,flag;

printf("\n Enter the N value :");

scanf("%d",&n);

int a[n]

printf("Enter the array Elements : |\n";

```

```

for( i=0;i<n;i++)
scanf("%d",&a[i]);
printf("Unique Elements:\n");
for( i=0;i<n;i++) {
flag =1;
for(j=0;j<n;j++) {
if(i!=j && a[i]==a[j]) {
flag=0
break;
}
}
if(flag)
printf("%d ",a[i]);
}
return 0;
}

```

Output:

Enter the N value : 6

Enter the array Elements :

15 22 15 88 99 22

Unique Elements:

88 99

5: Insert into Sorted Array

```
#include <stdio.h>
```

```
int main() {
```

```

int n,i,pos, value;

printf("\n Enter the N value:");

scanf("%d",&n);

}

int a[100];

printf("Enter the array Elements : \n");

for(i=0;i<n;i++)

scanf("%d",&a[i]);

printf("Enter value: ");

scanf("%d",&value);

for( i=0;i<n;i++) {

}

if(a[i]>value)

break;

pos =i

for(i=n;i>pos;i--)

a[i]=a[i-1];

a[pos]=value;

n++;

for(i=0;i<n;i++)

printf("%d ",a[i]);

return 0;

```

Output:

Enter the N value: 5

Enter the array Elements :

10 20 30 50 60

Enter value: 40

10 20 30 40 50 60

6: Insert into Unsorted Array

```
#include <stdio.h>

int main() {
    int n,i,pos, value;
    printf("\n Enter the N value :");
    scanf("%d",&n);
    int a[100];
    printf("Enter the array Elements : \n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    printf("Enter position: ");
    scanf("%d",&pos);
    printf("Enter value: ");
    scanf("%d",&value);
    for(i=n;i>=pos;i--)
        a[i]=a[i-1];
    a[pos-1]=value;
    n++;
    for(i=0;i<n;i++)
        printf("%d ",a[i]);
    return 0;
}
```

Output:

Enter the N value : 5

Enter the array Elements :

99 11 44 22 77

Enter position: 3

Enter value: 55

99 11 55 44 22 77

7: Matrix Multiplication

```
#include <stdio.h>

int main()
{
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d", &r1, &c1);
    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d", &r2, &c2);

    if (c1 != r2)
    {
        printf("Matrix multiplication is not possible.");
        return 0;
    }

    int a[r1][c1], b[r2][c2] c[r1][c2];

    printf("Enter first matrix elements:\n");
    for (i=0; i < r1; i++)
    {
        for (j=0; j<c1; j++)
```



```
{
scanf("%d", &a[i][j]);
}
}

printf("Enter second matrix elements:\n");
for (i=0; i<r2; i++)
{
for (j=0 j<c2; j++)
{
scanf("%d", &b[i][j]);
}
}

for (i=0; i < rl; i++)
{
for (j=0 j<c2; j++)
{
c[i][j] = 0;
for (k=0; k < cl; k++)
{
c[i][j] += a[i][k] * b[k][j];
}
}
}

printf("Product Matrix:\n");
for ( i=0 i < rl; i++)
{
```

```
for (j=0; j<c2; j++)
{
printf("%d ", c[i][j]);
}
printf("\n");
}
return 0;
}
```

Output:

Enter rows and columns of first matrix: 3 3

Enter rows and columns of second matrix: 3 3

Enter first matrix elements:

1 2 0

0 1 1

2 0 1

Enter second matrix elements:

1 1 2

2 1 1

1 2 1

Product Matrix:

5 3 4

3 3 2

3 4 5

8: Matrix Transpose

```
#include <stdio.h>
```

```
int main()
{
int r, c, i, j;

printf("Enter number of rows and columns: ");

scanf("%d%d", &r, &c);

int a[r][c]

printf("Enter matrix elements:\n");

for (i=0; i < r; i++)
{
for (j=0; j < c; j++)
{
scanf("%d", &a[i][j]);
}
}

printf("Transpose Matrix:\n");

for (i=0; i < c; i++)
{
for (j=0; j < r; j++)
{
printf("%d ", a[j][i]);
}

printf("\n");
}

return 0;
}
```

Output:

Enter number of rows and columns: 3 3

Enter matrix elements:

10 20 30

40 50 60

70 80 90

Transpose Matrix:

10 40 70

20 50 80

30 60 90

9: Right Diagonal Sum

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int n, i, j, sum=0;
```

```
printf("Enter the order of matrix: ");
```

```
scanf("%d", &n);
```

```
int a[n][n];
```

```
printf("Enter matrix elements:\n");
```

```
for (i=0;i<n;i++)
```

```
{
```

```
for r(j=0;j<n;j++)
```

```
{
```

```
scanf("%d", &a[i][j]);
```

```
}
```

```
}
```

```
for (i=0;i<n;i++)
{
sum+=a[i][n-1-i];
}
printf("Right diagonal sum = %d", sum);
return 0;
}
```

Output:

Enter the order of matrix: 3

Enter matrix elements:

2 4 6

1 5 9

8 3 7

Right diagonal sum = 19

10: Check Identity Matrix

```
#include <stdio.h>

int main()
{
int n, i, j, flag=1;
printf("Enter the order of matrix: ");
scanf("%d", &n);
int a[n][n];
printf("Enter matrix elements:\n");
for (i = 0; i < n; i++)
{
```

```
for (j = 0; j < n; j++)
{
scanf("%d", &a[i][j]);
}
}
for (i=0;i<n;i++)
{
for (j = 0; j < n; j++)
}
{
if (i==j&&a[i][j]!=1)
{
flag=0;
}
if (i != j && a[i][j] != 0)
{
flag=0;
}
}
}
if (flag==1)
printf("It is an Identity Matrix.");
else
printf("It is not an Identity Matrix.");
return 0;
```

Output:

Enter the order of matrix: 3

Enter matrix elements:

1 0 0

0 1 0

0 0 1

It is an Identity Matrix.

11: Count Vowels, Consonants, Digits and White Spaces

```
#include <stdio.h>

int main()
{
    char str[100];

    int i, vowels =0, consonants =0, digits =0, spaces =0;

    printf("Enter a string: ");

    scanf("%s", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if ((str[i]>='A'&str[i]<='Z')||(str[i]>='a'&str[i]<='z'))
        {
            char ch = str[i];

            if (ch >= 'A' && ch <= 'Z')
            {
                ch=ch+32;

                if (ch=='a' ||ch=='e' ||ch=='i' ||ch=='o' ||ch=='u')
                {
                    vowels++;
                }
            }
        }
    }
}
```

```
else
consonants++;
else if (str[i]>='&&str[i]<='9')
{
digits++;
}
else if (str[i]=="")
{
spaces++;
}
}

printf("Vowels = %d\n", vowels);
printf("Consonants = %d\n" consonants);
printf("Digits = %d\n", digits);
printf("White Spaces = %d\n" spaces);

return 0;
```

Output:

Enter a string: Data Structures 2026

Vowels = 5

Consonants = 9

Digits = 4

White Spaces = 2

12: Frequency of a Character

```
#include <stdio.1>
```

```
#include <stdio.h>
```



```
int main()
{
char str[100], ch;
int i, count =0;
printf("Enter a string: ");
scanf("%s", str);
printf("Enter the character: ");
scanf("%c", &ch);
{
for( i=0; str[i] != '\0'; i++)
if(str[i]==ch)
{
count++;
}
}
printf("Frequency of %c=%d", ch, count);
return 0;
}
```

Output:

Enter a string: programming

Enter the character: g

Frequency of g=2

13: Count Words in a String

```
#include <stdio.h>
```

```
int main()
```

```
{  
char str[100];  
int i, words =1  
printf("Enter a string: ");  
scanf("%s", str);  
  
{  
for(i=0;str[i]!='\0';i++)  
if(str[i]=='  
{  
words++;  
}  
}  
  
printf("Number of words = %d", words);  
return 0;  
}
```

Output:

Enter a string: C programming language assignment

Number of words = 4

14: Sort String Array

```
#include <stdio.h>  
  
#include <string.h>  
  
int main()  
{  
int n, i, j;  
  
char str[20][50], temp[50];
```

```
printf("Enter the number of strings: ");
scanf("%d", &n);
printf("Enter the strings:\n");
for(i = 0; i < n; i++)
{
scanf(" %[^\\n]", str[i]);
}
for(i = 0; i < n - 1; i++)
{
for(j = i + 1; j < n; j++)
{
}
}
if(strcmp(str[i], str[j]) > 0)
{
strcpy(temp, str[i])
strcpy(str[i], str[j]);
strcpy(str[j], temp);
;
}
printf("Strings in Sorted Order:\n");
for(i = 0; i < n; i++)
{
printf("%s\\n", str[i]);
}
return 0;
```

```
}
```

Output:

Enter the number of strings: 4

Enter the strings:

Zebra

Apple

Mango

Banana

Strings in Sorted Order:

Apple

Banana

Mango

Zebra

15: Toggle Case of a Sentence

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
char str[100];
```

```
int i;
```

```
printf("Enter a sentence: ");
```

```
scanf("% [^\n]", str);
```

```
or(i=0;str[i]!="0";i++)
```

```
if(str[i]>='A'&str[i]<='Z')
```

```
{
```

```
{
```

```
str[i]=str[i]+32;
}
else if(str[i]>=a'&&str[i]<=z')
{
str[i]=str[i]-32;
}
}
}

printf("Sentence after toggling:\n%s", str);
return 0;
```

Output:

Enter a sentence: CPrOgRaMmInG

Sentence after toggling:

cpRoGrAmMiNg